

# Automatic extraction of semantic AOIs from heterogeneous SVG visualizations

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**Abstract**

TODO

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## I. Introduction, Hypotheses & Motivations

Given a predefined set of Scalable Vector Graphics visualizations (maps, treemaps, scatterplots, ETC...), this project aims to build a set of tools capable of extracting relevant information, mainly position and semantic, of the Areas Of Interest present in the SVG. This project is initially part of a broader initiative to extract eye-tracking data from these same SVGs, the ultimate goal being to correlate gaze patterns with AOIs in order to better interpret the whole process one would use to understand various visualizations. With the aim of generalizing the approach, this project seeks to extract semantic information from SVG visualizations beyond the original — and already very verbose — dataset.

As discussed in the introduction, the original dataset was composed of very verbose SVGs. By that, we mean that these SVGs were greatly annotated, with many attributes, such as `ids` and `classes`, already having relevant semantic information. That is the first hypothesis made, parsing these attributes properly would already give a good idea of the semantics of the elements making up the visualizations. Combined with the tree structure of the SVG format itself [1], we initially thought that the original dataset could practically be fully interpreted this way. After discussing with our supervisor, we understood that the SVGs of the dataset had already been processed manually a lot for the sake of research, our *naïve* approach was thus not satisfactory, and needed a more general strategy. We then made the hypothesis that the semantics of the AOIs could come from a *broader* context, which will be discussed in Section II.

Since our goal is also to extract positional information about the AOIs, we also needed a strategy to compute *bounding boxes*, in order to have a mapping of the semantic information in the space of the SVG. Even though the geometry is not trivial, the tree structure of the SVG combined with the vector description of the shapes made the computation of precise spatial boundaries a systematic, albeit complex, geometric problem. By traversing the document tree and translating the raw vector paths and affine transformations into exploitable geometric objects, we hypothesized that we could accurately compute these bounding boxes, regardless of the shape's complexity. This spatial mapping is a critical prerequisite: without a reliable system to map an element's extracted semantics to its exact 2D coordinates, correlating the data with eye-tracking logs would be impossible.

In the remainder of this report, we will detail our implementation pipeline, from the initial parsing of the SVG Document Object Model (DOM) to the geometric processing and generalized semantic extraction. We will then discuss the technical challenges encountered—particularly regarding non-standard SVG structures and complex composite shapes—before evaluating our findings and our alignment with the project's initial objectives.

## II. Implementation Process

In this section, we discuss both the technical and managerial aspect of our project's pipeline implementation.

## **|| II.a. Development Roles & Assignments**

## **|| II.b. Technical Strategies**

## **| III. Challenges & Findings**

## **| IV. Results**

## **| V. Discussion**

## **| VI. AI Usage**

## **| References**

- [1] World Wide Web Consortium (W3C), “Scalable Vector Graphics (SVG) Specification.” [Online]. Available: <https://www.w3.org/TR/SVG/>